CSE 108: Computer Programming Lab **08.05.2026**

Lab #10 - Structs and Unions

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Scenario

In this lab, you will complete a C program that calculates the total monthly cost of different digital subscription services.
The program works with three subscription types:

- Music Streaming
- Cloud Storage
- Online Course Platform

Each subscription type requires different information. Since only one subscription type is processed at a time, the program must use a `union` to store the selected subscription data.
A C source file is provided to you. It contains the general program structure, the `main` function, and missing parts marked with `TODO` comments. Your task is to complete these missing parts.

Learning Objectives

By the end of this lab, you should be able to:

- define and use `struct` types,
- define and use a `union`,
- access nested members using the dot operator,
- pass structures to functions,
- return structures from functions.

Required Data Structures

You must define three different structs:

- `music_t`
- `cloud_t`
- `course_t`

All members of these structs must be defined as `double` variables.

```
music_t:
monthly_fee
extra_hours
extra_hour_fee
total_cost

cloud_t:
base_fee
extra_gb
gb_fee
total_cost

course_t:
monthly_fee
number_of_courses
course_fee
total_cost
```

Union Definition

Define a union named `subscription_data_t`. Inside the union, include the following members:

```
music_t music;
cloud_t cloud;
course_t course;
```

This union stores only one type of subscription data at a time.

General Subscription Struct

Define a struct named `subscription_t`. Inside this struct, store:

```
char type;
subscription_data_t data;
```

The `type` member represents the selected subscription type:

- `'M'` for music streaming
- `'C'` for cloud storage
- `'O'` for online course platform
- `'Q'` for quit

Functions to Implement

You must implement the following two functions in the provided C file:

```
subscription_t get_subscription_info(void);

subscription_t compute_total_cost(subscription_t subscription);
```

`get_subscription_info` must ask the user to enter the subscription type and the required values for that type. If the user enters a character other than 'M', 'C', or 'O', the function must store 'Q' as the type.

`compute_total_cost` must calculate and print the total monthly cost according to the selected subscription type.

Cost Formulas

```
Cost of Music Streaming = monthly_fee + extra_hours * extra_hour_fee

Cost of Cloud Storage = base_fee + extra_gb * gb_fee

Cost of Online Course Platform = monthly_fee + number_of_courses * course_fee
```

The calculated value must be stored in the appropriate `total_cost` member of the selected union member.

Sample Run

A possible run of the program is shown below.

```
Digital Subscription Cost Computation Program

Enter subscription type (M for music, C for cloud, O for online course, Q to quit): C
Enter base fee: 100
Enter extra storage in GB: 50
Enter fee per extra GB: 1.5
Total cost for cloud storage: 175.00

Enter subscription type (M for music, C for cloud, O for online course, Q to quit): M
Enter monthly fee: 60
Enter extra listening hours: 10
Enter fee per extra hour: 0.5
Total cost for music streaming: 65.00

Enter subscription type (M for music, C for cloud, O for online course, Q to quit): O
Enter monthly fee: 150
Enter number of additional paid courses: 2
Enter fee per additional course: 40
Total cost for online course platform: 230.00

Enter subscription type (M for music, C for cloud, O for online course, Q to quit): Q
```

Important Notes

- Do not change the given `main` function.
- Complete only the missing parts marked with `TODO`.
- Use the `type` member to decide which union member is valid.
- Print the calculated total cost inside `compute_total_cost`.
- Make sure your program compiles and runs correctly.